SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation?

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Abstract

Dynamic manipulation requires a robot to coordinate contact with a scene that continues to change between observation and execution. This raises a question beyond predicting more of the scene: which historical and future information actually improves the action? We study history encoding, future representation, and world–action coupling in a common dynamic-manipulation framework. Our guiding hypothesis is that prediction should preserve target motion and task-relevant context at the granularity required by interaction. A completed fifteen-condition future-supervision study on fourteen simulated tasks shows why this hypothesis requires controlled testing. On 1,260 paired evaluation scenes, the highest point estimates are 64.13% and 64.05%, compared with 62.78% without future supervision, with no clear overall advantage. Dense 2-D displacement supervision reaches 59.21% and is worse under the reported multiplicity correction. These results use privileged current object state and motion history, one training seed, and an auxiliary prediction path. History and combination studies further separate information access from representation choice. We specify the remaining backbone, online information-flow, pretraining, and external evaluations that will determine the final SE(3)-WAM instance. The present evidence supports conditional design decisions, while leaving the value of deployable online foresight to those comparisons.

1. Introduction

A moving object can leave its observed position before a robot reaches it. The same problem arises when a receptacle moves during release or an obstacle changes the available approach corridor. Dynamic manipulation therefore requires spatial accuracy at the right time. Faster inference and more responsive execution reduce observation age [30], while temporal observations help distinguish motions that look similar in a single image [8, 20]. Together, these demands motivate a practical question: *what should a robot remember and predict so that its action remains useful when interaction occurs?*

The answer depends on the role of information in the task. Target translation and rotation determine where and how to make contact. A moving occluder or obstacle matters through its effect on visibility, reachability, or collision. Static geometry can remain essential even though it has no motion to predict. This suggests organizing scene evolution by interaction needs: preserve fine target information when contact requires it, and represent surrounding changes at sufficient resolution to support the decision. Both a fixed target/context decomposition and a shared representation with task-dependent readout can implement this idea. Their relative value is an empirical question.

World–action models provide a framework for learning scene evolution alongside robot actions [33, 34]. However, a future objective changes several things at once: the supervised variables, the gradients shaping shared features, and sometimes the computation used before acting. The action may benefit from auxiliary supervision even when no forecast is generated at deployment [36]. Conversely, a geometrically accurate forecast may add no useful information to an already strong state-conditioned policy. Prediction error, action improvement, and online forecast use must therefore be measured separately.

We organize the SE(3)-WAM Study around three connected questions. **Q1** asks how a finite history should encode motion. **Q2** asks which future variables, regions, and resolutions support action. **Q3** asks how world and action computation should share parameters and exchange information. The study first establishes a common reference, compares history and future choices, and then selects a backbone and its information flow. Targeted cross-comparisons revisit earlier choices before the complete design is used for robot-data pretraining. This ordering exposes interactions between design decisions without requiring an exhaustive Cartesian product.

The completed evidence already constrains the story. Fifteen future-supervision conditions have been trained and evaluated on a 1,260-scene complement of the development

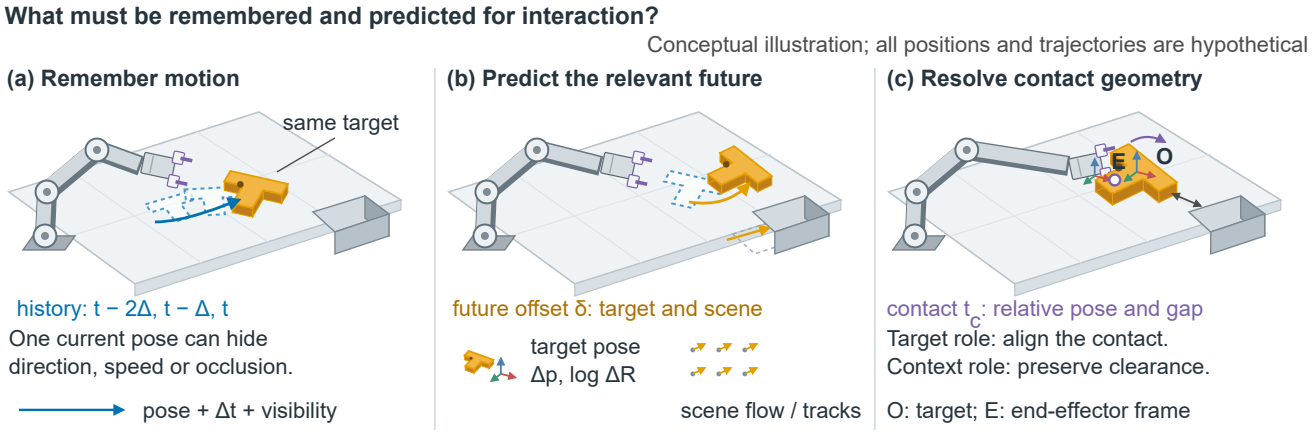


Figure 1. What information matters at interaction time? (a) History disambiguates motion hidden by a single target pose. (b) Future target pose and scene motion describe complementary physical quantities. (c) Contact requires end-effector/target alignment and clearance from context. The same asymmetric target and robot connect the panels. All trajectories are conceptual illustrations of the research question.

set. Sparse scene or target trajectories give the highest point estimates, but neither clearly exceeds the no-future reference overall. Dense 2-D future displacement decreases success under the same comparison. A subsequent twelve-condition combination screen selects a candidate with 66.59% success on that complement, while its exact matched controls remain incomplete. An adapted DreamVLA reference reaches 79.60% under its own recipe. These observations motivate investigating how information is encoded and used, rather than treating the addition of future targets as an established benefit.

Our contributions at this stage are:

- A dynamic-manipulation study protocol that separates task outcomes, available information, training supervision, and execution conditions.
- A completed comparison of fifteen future-supervision conditions, with paired results and adverse findings, complemented by history and combination studies that identify the next controlled contrasts.
- A unified design and evaluation framework for selecting SE(3)-WAM through history, future-representation, backbone, and information-flow comparisons, followed by a specified pretraining and transfer study.

The supplement provides complete configurations, results, and remaining protocols.

2. Related Work

Dynamic manipulation and physical time. Motion-aware grasping connects prediction to reachability and interception [1]. DOMINO/PUMA couples a dynamic benchmark with historical flow and object-future supervision [8]; DynamicManip expands static demonstrations into dynamic data [15]. DynamicVLA emphasizes continuous inference and action streaming [30], while DynamicWAM conditions world–action learning on motion [20]. Temporal feature prediction [39], AHEAD's predictive latents [24], and ForeTime-VLA's future-token and event-time supervision [21] offer different ways to anticipate interaction. RTC addresses continuity under delayed chunk generation [2]; Chronos Policy explicitly represents observation and execution times [26]. These works motivate evaluating prediction together with action age, rather than identifying a simulator control interval with real-time operation.

History and object-centered information. TraceVLA encodes temporal traces as visual prompts [41]; HiF-VLA organizes past and future dynamics through motion representations [16]. PACT-WAM allocates fewer spatial tokens to older observations [18], and Long-WAM studies long context together with causal video pretraining [11]. Thus a history encoder's value depends on both temporal support and the training that makes that information usable. DOCIR separates manipulated objects, obstacles, and the robot [7]; OA-WAM makes objects addressable within world–action modeling [19]. Our Q1 comparisons distinguish current localization, privileged motion, and visual history before comparing encodings. Role assignment is an experimental factor, not a presumed optimum.

Which future should a policy predict? SE3-Nets models scene motion through rigid transforms [3]; SE3-Pose-Nets uses structured pose dynamics for control [4]. DSR-Net accumulates a volumetric object representation over interactions [32], while PointWorld and PhysMani develop 3-D world representations for manipulation [10, 37]. ATM learns point trajectories from video [28]; Im2Flow2Act uses generated object flow as a policy interface [31]. AffordTrajDP

propagates contact affordances with object pose [29], and FlowWAM uses flow in policy and world-model modes [5]. These interfaces differ in whether they describe desired motion, action-conditioned consequences, or externally moving objects. DreamVLA and DreamWAM add complementary visual, geometric, and semantic targets [35, 38]; DC-WAM instead emphasizes dynamics within RGB supervision [12]. DELE-w0.5 uses compact future latents [14]. Q2 places related target families under one action path to measure their conditional training value; its pose output does not imply a new rigid-motion parameterization.

World–action coupling and experimental attribution. VPP conditions action learning on predictive video features [9], whereas DreamZero jointly models world and action dynamics [33]. Fast-WAM separates training benefits from test-time imagination [36]. Two distinct OpenWAM projects provide relevant design studies: Wang et al. investigate modular pretraining and release the Alpha foundation used by our adapted baseline [27]; Yu et al. compare composable world–action programs in a common framework [34]. Neither is the pretrained source of our freshly initialized reference D backbone. Q3 separates shared-parameter learning, cross-stream access, and generation order, then revisits representation choices. ACT, Diffusion Policy, and $\pi_{0.5}$ supply complementary system references [6, 22, 40]; DROID supplies the planned real-robot pretraining data [13]. Native system comparisons and matched design contrasts answer different questions and are reported separately.

3. Problem and Evaluation Protocol

Interaction in physical time. Let $\mathcal{H}_t = \{(o_i, s_i, \tau_i)\}_{i \le t}$ denote the available finite history of observations, robot state, and timestamps. A policy produces an action chunk $A_t = (a_{t,0}, \dots, a_{t,H_a-1})$. If the latest image was captured at τ_t^s and the first command is applied after delay ℓ_t, its application times are

$$\tau_{t,j}^{\text{apply}} = \tau_t^s + \ell_t + j\Delta_a. \tag{1}$$

The relevant future is defined relative to these physical times and the task's contact event. Forecast horizons, action intervals, and computation time are different quantities. In the completed representation screen, labels use eight future samples from 50 to 400 ms, while evaluation uses synchronous policy queries and executes one action per replan. Real computation delay is not injected into the physical evolution by those synchronous results.

Tasks and data. The reported experiments use the migrated DB0 v0.6 data and the fourteen-task evaluation suite. The suite covers moving-target grasping, orientation-sensitive grasping, physical occlusion, moving-receptacle placement, phase-constrained passage, free rolling/sliding motion, and post-impact adaptation. The simulation infrastructure uses MuJoCo and robosuite [25, 42]. Task success is determined by each native evaluator's physical outcome; all fourteen tasks receive equal weight. The common demonstration pool contains 11,200 trajectories, 800 per task. P0 motion-only prediction and later fresh-data production are separate from this learning/evaluation record. Appendix A describes the task and split identities used here.

Information tracks. The future-supervision screen uses a current fixed-camera RGB image, exact depth/camera geometry, an eight-sample robot-state history, and privileged current target pose plus seven intervals of target kinematics. We call this the *object-motion information track*. It isolates future objectives when substantial target motion information is already available. The current-pose history study removes privileged interval kinematics and asks whether visual history supplies useful additional information. Both tracks still use privileged current target pose. A deployable visual system must estimate such quantities from allowed sensors and is evaluated separately. Current-only RGB with robot-state history is denoted H0V; it retains memory.

Evaluation partitions. The fixed140 development set contains ten scenes per task. The disjoint 1,260-scene complement contains ninety per task, giving 1,400 scenes together. The complement is independent of fixed140 membership but belongs to the same validation distribution and has now informed development. The combined 1,400-scene score includes development scenes and is used for combination selection; it is not a new out-of-distribution test. We preserve these identities in every table. Checkpoints in the completed future screen use the fixed fourth-epoch endpoint and one fresh training seed. Paired comparisons share scenes and specified rollout seeds; scenario uncertainty does not include between-training-seed variation.

Outcomes and comparison scope. For $K = 14$ tasks with n_k episodes each, the primary score is

$$\mathrm{SR}(m) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{i=1}^{n_k} y_{mki}, \qquad y_{mki} \in \{0,1\}. \tag{2}$$

Equal per-task counts make pooled and task-balanced SR identical here. We report counts as well as percentages. Paired differences use the same episodes and the uncertainty procedure attached to the source analysis. The completed future comparison corrects its twenty-seven overall contrasts as one family. Per-task findings are exploratory.

Matched internal studies share observation fields, training exposure, action interface, and execution settings. The external-method adaptation table instead compares complete

systems with different pretraining, training budgets, and decoders. It records a useful performance reference without identifying a particular architecture's effect. Contact-time and relative-pose errors, safety outcomes, and measured action age are reserved for the mechanism and deployment studies where their measurement interfaces are available. Missing readouts are not assigned zero. Representation-specific errors retain their native units; a latent loss is not ranked numerically against a metric pose error.

4. The SE(3)-WAM Design Study

The study represents a design by (H, Z, B, G): a history encoder H, future targets Z, backbone B, and information-flow program G. Figure 2 shows their roles. The completed screen changes Z under a fixed reference; subsequent stages examine H, B, and G and revisit retained alternatives. A single component's point-estimate ranking does not determine the final composition.

4.1. Q1: Information Access and History Encoding

The encoder maps timestamped history into conditioning tokens, $X_t = H(\mathcal{H}_t)$. Its design includes the physical window, temporal spacing, visibility, and motion coordinates. Current RGB, stacked RGB windows, codec motion vectors, optical flow, latent history, and per-frame query compression provide different descriptions of the same recent evolution. Encoding and information access are separate factors: supplying a ground-truth velocity changes what is known, while reorganizing the same visual samples changes how it is represented.

The current-pose study holds current target pose and eight robot-state samples fixed, removes target-history kinematics, and compares eight visual-history conditions plus a freshly trained GT-history reference. The object-motion track instead supplies seven intervals of target kinematics. This separation tests the conditional value of visual history rather than conflating temporal encoding with target localization. Window length and timestamp spacing are recorded independently of token count. A motion displacement also carries its time interval; a trajectory contains visibility rather than treating an absent point as stationary.

The diagnostic program complements task success with motion probes, matched-current history substitutions, occlusion, and sampling-gap tests. A probe measures what is decodable. A history substitution tests whether a policy responds to temporal information. Useful memory additionally requires better action outcomes under valid matched conditions. The necessary-current matching and closed-loop diagnostics remain part of the unfinished Q1 study.

4.2. Q2: Predicting Interaction-Relevant Variables

Q2 separates *scope* (target, context, or both), *variables* (pose, motion, visual features), and *granularity* (spatial and temporal allocation). The completed comparison includes no future target, scene-flow only, target-pose only, their combination, alternative scene/target descriptions, and additional future supervision. Figure 3 gives the central contrasts; Appendix D provides all fifteen conditions. Scene flow also covers target regions, so the scene-flow-only condition is not a context-only control. Pure role separation and allocation require additional matched comparisons.

For the rigid-target condition, let (p_t, R_t) be the current target pose in the fixed metric frame. At physical offset $\delta_h = 0.05h$ seconds, $h = 1, \ldots, 8$, the supervised variables are

$$d_h = p_{t+\delta_h} - p_t, \qquad \phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee. \quad (3)$$

Predicted poses reconstruct as $\hat{p}_{t+\delta_h} = p_t + \hat{d}_h$ and $\hat{R}_{t+\delta_h} = \mathrm{Exp}(\hat{\phi}_h^\wedge) R_t$. Translation and left-relative rotation are parameterized separately. The pose readout pools sixteen meta tokens and predicts eight cumulative transforms. Its valid-target loss combines squared translation error with 0.1 times squared rotation geodesic error. This geometric parameterization does not impose equivariance on the network.

Scene-flow supervision describes metric cumulative point displacement. Alternatives describe dense 2-D displacement, codec block motion, sparse scene trajectories with visibility, or future video latents. Target alternatives use pooled DINO features or target trajectories. The added-supervision conditions predict depth, DINO features, dynamic-region RGB, or their combination. Their labels have different units, support, and dimensionality. Replacing scene flow with dense 2-D motion consequently tests a representation package, not dimensionality alone. Likewise, pose plus sparse tracks is not by itself a matched proof that fine-target/coarse-context allocation is optimal.

The completed study uses these quantities as auxiliary training targets. Future labels and validity masks remain in their supervision branches; neither is an action input. Deterministic readouts and generated variables have distinct interfaces. To compare uniform and heterogeneous allocation directly, the remaining study must hold role information, temporal support, and an applicable token/compute budget common, then vary the allocation within a compatible representation. A target mask used to construct a training label does not establish deployable target assignment.

4.3. Reference Training and Action Generation

The common Q2 reference uses a four-layer transformer of width 768 with eight attention heads. A frozen DINOv3 ViT-L/16 encoder [23] provides 196 tokens from a 224×224 current image. The D backbone shares attention projections and uses separate feed-forward networks for world/meta, flow, and action streams. Action attends to meta and action tokens; it does not attend to predicted flow or flow hidden

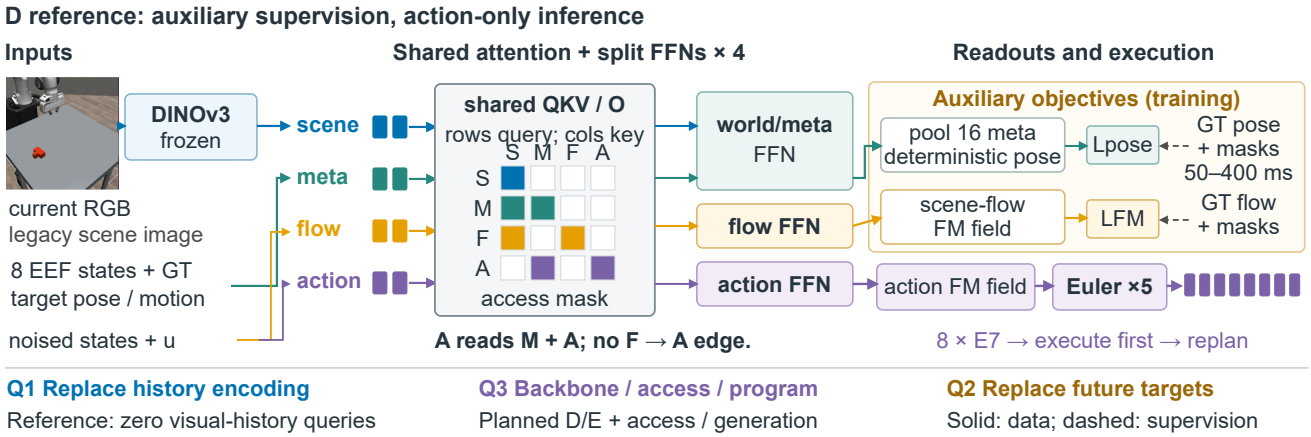


Figure 2. Implemented reference D, with P03's pose and scene-flow targets shown. Current RGB, robot history, and privileged target state form conditioning tokens. The mask gives query-to-key access for scene (S), meta (M), flow (F), and action (A). Within each of four layers, the streams share attention projections and use stream-specific FFNs. Pose is a deterministic readout; scene flow and action use flow-matching fields. Future labels and masks provide training supervision (dashed). Deployment solves only action with five Euler steps, executes the first E7 command, and replans. Depth and camera geometry are omitted for clarity; conditioning details are in Appendix B. Q2 changes the enabled targets; Q1/Q3 change history and computation.

states. Future objectives can still shape shared trainable features through their gradients.

Action generation uses flow matching [17]. With normalized action target a_1, source $a_0 \sim \mathcal{N}(0, I)$, and interpolation time $u \in [0, 1]$,

$$a_u = (1-u)a_0 + ua_1, \quad \mathcal{L}_A = \mathbb{E}\|v_\theta(a_u, u, X_t) - (a_1 - a_0)\|^2. \tag{4}$$

The complete objective adds only the enabled future losses, $\mathcal{L} = \mathcal{L}_A + \sum_{r \in Z} \lambda_r \mathcal{L}_r$. Masks restrict each loss to its valid support. A missing head contributes no measurement; raw total losses with different terms are not compared as one quality metric. Appendix B specifies the reference source distributions, coordinates, and implementation settings.

The action chunk contains eight seven-dimensional commands: translation increment, left-relative rotation vector, and a gripper command. At deployment, five Euler steps generate the action chunk, its first command is executed, and the policy replans. Conditioning caching is enabled; temporal and velocity caching are disabled. The training/evaluation translation bounds are respectively 0.05/0.10 m per axis. Only the action stream is solved online in the completed Q2 screen.

4.4. Q3: Backbone and Information Flow

The next backbone comparison holds selected H/Z and the reference action-only deployment program G_0 fixed. It compares D's stream-specific feed-forward organization with an adapted E world/action parameter organization. Both are freshly trained; the proposed active-backbone parameter difference is bounded by approximately 5%. This E is distinct from the object-input condition called E in earlier experiments and from the older native two-column E implementation. It is a planned matched backbone comparison, with no completed GPU result.

The subsequent seven-condition information-flow study holds $H/Z/B$ fixed: R0 removes the studied future loss; R1 retains it without cross-stream edges; R2 allows future-to-action access; R3 allows action-to-future access; R4 allows both; R5 generates future then action; R6 generates action then future. R1 also compares action-only and joint deployment of the same checkpoint. These contrasts distinguish supervision, information access, and generation programs. The sequential conditions use clean ground-truth first-stage values for second-stage training and predicted values at deployment, so their difference includes teacher forcing and its deployment mismatch.

The current generative interface is defined for scene flow. If representation selection retains sparse tracks or video latents, their generated states, normalization, noise, and loss must be specified before the information-flow comparison. Five joint evaluations and five-plus-five sequential evaluations match per-modality steps but not total compute. The planned comparison therefore reports actual calls, parameters, and same-device latency. Online substitutions of future, including oracle and valid swaps, are used only after a replaceable forecast interface exists.

5. Experiments

We report completed system references, future-supervision comparisons, combination selection, and the available history results. The backbone, information-flow, pretraining, and external studies are specified in Sec. 5.6. Tables distin-

Table 1. System references on the 1,260-scene complement (fourteen tasks, ninety scenes each). Training seed is 0. Methods retain their adapted input, initialization, and execution pipelines; budgets and fixed140 counts are in Appendix B. Scores compare complete systems, not matched architectures. The lower block receives GT target pose and interval kinematics.

System	Initialization / study condition	Successes /1,260	SR (%)
Adapted published systems			
ACT [40]	From scratch	424	33.65
Diffusion Policy [6]	From scratch	432	34.29
$\pi_{0.5}$ [22]	Official `pi05_base`	751	59.60
DynamicVLA [30]	Adapted native pipeline	363	28.81
OpenWAM-Alpha [27]	Official Alpha foundation	708	56.19
DynamicWAM [20]	Official full initialization	670	53.17
DreamVLA [38]: 20 epochs	Independent 20-epoch training schedule	1,000	79.37
40 epochs	Independent 40-epoch training schedule	1,003	79.60
Controlled-study systems with privileged target information			
P00: no future supervision	Current image; frozen DINO-L	791	62.78
P03: scene flow + pose	Current image; frozen DINO-L	786	62.38
I: scene tracks + pose	Full visual window; CNN; target RGB reconstruction	839	66.59

guish measured values from pending cells.

5.1. System References and Training Conditions

Table 1 places the study against adapted action and world–action models on the same fourteen-task evaluation scenes. The two DreamVLA schedules reach 1,000/1,260 and 1,003/1,260 successes. The study's combination candidate I reaches 839/1,260, and the no-future reference reaches 791/1,260. These are system-level observations. Candidate I and the reference have privileged target information; the adapted methods differ in initialization, training exposure, observation processing, and action execution. The table therefore motivates improving the current system while retaining the stronger reference, without attributing the gap to future representation.

Within the fifteen-condition future screen, all task networks start fresh with training seed 0 and run four epochs, 40,104 updates, and 641,664 sampled chunks at global batch 16. They use the fixed fourth-epoch checkpoint. Training is distributed over eight or sixteen devices with the same global batch; topology histories are retained in Appendix B. A single trained endpoint per condition supports paired scene comparisons but does not measure training-seed robustness. Current pose, motion history, action objective, and one-step receding-horizon execution are held common across this screen.

5.2. Q2: Which Future Supervision Helps?

Adding scene-flow and target-pose objectives. Figure 3 places the core contrasts in the full comparison. The no-future reference P00 succeeds in 791 scenes. Scene-flow P01, target-pose P02, and their combination P03 succeed in 782, 789, and 786 scenes, respectively. The combined objective does not show an overall improvement over either component or the reference in this setting. This result concerns auxiliary training with privileged motion input; online forecast consumption is a separate contrast.

Changing the predicted variables. Sparse scene tracks with target pose (P07) reach 807/1,260, and scene flow with target tracks (P09) reaches 808/1,260. Relative to P00, P07 gains 85 scenes and loses 69; P09 gains 91 and loses 74. Their raw two-sided exact McNemar p values are 0.227 and 0.213. The small net gains do not establish a reliable overall advantage. Non-rejection also does not establish equivalence: P12, for example, has the same total as P00 but exchanges 73 successes and 73 failures.

Dense 2-D future displacement with pose (P05) reaches 746/1,260, a decrease of 3.57 percentage points from P00 and 3.17 points from P03. The corresponding Bonferroni-adjusted p values over twenty-seven overall contrasts are 0.00844 and 0.01877. The comparison supports an adverse effect for this representation condition under the reported recipe. It does not isolate a universal 2-D versus 3-D ordering: P07 and P09 also use 2-D trajectories, with different support and target definitions.

Task variation and failure evidence. All fifteen conditions obtain zero successes on rigid-motion receptacle placement (`t2_se3`). On full constrained passage (`t3_full`), P00 and P06 obtain 3/90 and 4/90, with zero for the remaining conditions. Prediction changes have therefore not removed these shared task failures. P05's loss relative to P03 includes 27 fewer successes on rolling-sphere capture; its native `tray_contact` failures increase from 40 to 67 in that task. P09 gains on moving placement and replanning but

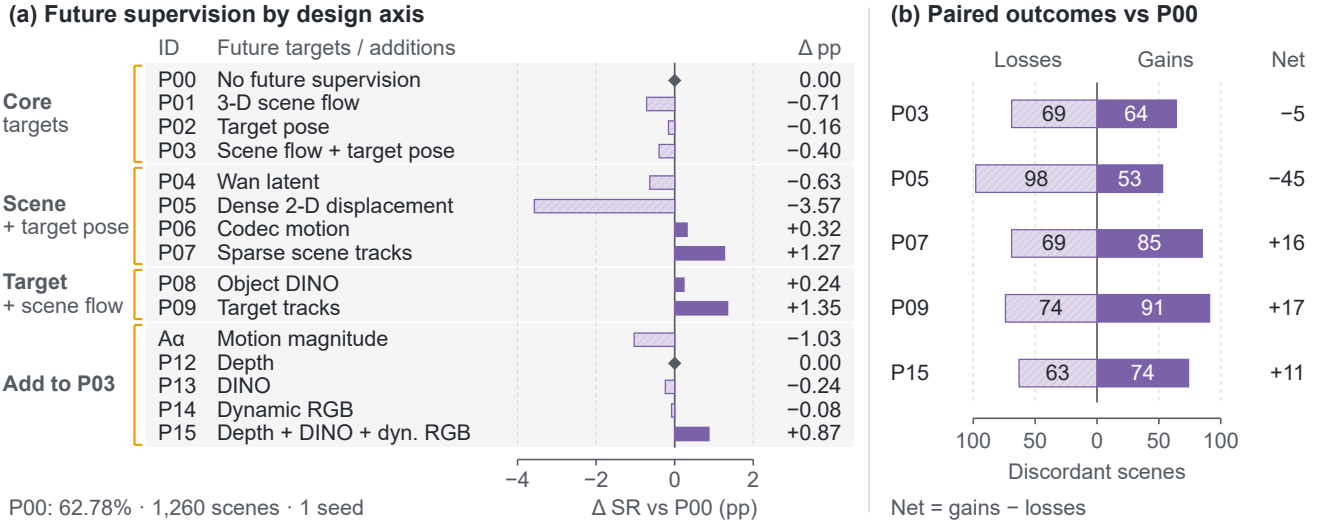


Figure 3. Future supervision at the fourth-epoch endpoint: one training seed, 1,260 paired scenes per condition (fourteen tasks, ninety each). (a) All fifteen conditions, grouped by the component changed; bars show success-rate differences from P00 (62.78%). (b) Selected contrasts expose scene-level changes: gain means P00 fails and the candidate succeeds; loss means the reverse. Net change is gain minus loss. P05 is adverse after the 27-contrast correction; P07/P09 do not establish an overall gain. No confidence intervals are shown. The separate fixed140 partition is in Appendix D.

loses on orientation-sensitive grasping. These observed failure codes and counts identify where to investigate; they do not identify a representation mechanism without an aligned intervention.

Development-set ordering. P01 has the highest fixed140 point estimate, while P09 has the highest 1,260-scene point estimate. The development-set ranking does not persist across the remaining validation scenes. Appendix D reports all conditions, paired gain/loss counts, and per-task results. The evidence favors retaining competitive alternatives for joint comparison instead of treating the first ranking as a final design principle.

5.3. Combining Future Targets with Visual Encoding

The twelve-condition combination study crosses P07/P09/P15 with DINO-S or a CNN encoder and with or without current-target RGB reconstruction. It uses a full visual window and privileged target-motion information. The predefined selection score is success over all 1,400 development/evaluation scenes. Candidate I, P07 with CNN and reconstruction, obtains 935/1,400 (66.79%), including 839/1,260 (66.59%) on the complement, and is selected for matched controls (Table 2).

Its 39-scene net gain over the same P07/CNN condition without reconstruction is concentrated: phase passage gains 44 scenes and orientation-sensitive grasping gains 17, while translational placement loses 10 and XYZ grasping loses 7. Both rigid-motion placement and full passage remain at zero for all twelve conditions. Thus the selection is useful for subsequent experimentation, but its overall score does not imply uniform task improvement. The matched P03 and P00 conditions using CNN and reconstruction are **[pending complete evaluation]**. Their outcomes are necessary to attribute any gain to the selected future target.

Table 2. Future–encoder combinations: successes /1,400 (the development selection score, including fixed140). Rec. denotes current-target RGB reconstruction. All conditions receive a full visual window and GT target motion. The P07/CNN/+rec. condition is selected for matched P00/P03 controls, which remain pending. Appendix D gives all candidate IDs and task counts.

	DINO-S		CNN	
Future supervision	No rec.	+rec.	No rec.	+rec.
P15: P03 + three targets	873	836	875	867
P07: scene tracks + pose	840	849	896	935
P09: flow + target tracks	820	830	829	870

5.4. Q1: Visual History Given Current Target Pose

Five of the nine current-pose study conditions have complete scores (Table 3). The current-image condition obtains 748/1,400 (53.43%); Window, Codec-MV, Flow, and PACT-Age obtain 662, 733, 720, and 634 successes. None of these completed history conditions exceeds the current-image control in point estimate. The remaining conditions and the fresh

Table 3. Visual history given current GT target pose and robot history. Historical target kinematics are removed. SR uses the 1,400-scene development total; Δ is relative to current image, in percentage points. Appendix C reports partition counts and four unfinished conditions separately.

Visual input	SR /1,400 (%)	Δ (pp)
Current image	53.43	0.00
Window	47.29	−6.14
Codec-MV	52.36	−1.07
Flow	51.43	−2.00
PACT-Age	45.29	−8.14

GT-history reference are pending, so the full Q1 comparison remains open.

This experiment preserves robot-state history and current GT target pose while removing target-history kinematics. It addresses visual motion information conditional on current localization. Its frozen DINO-L encoder, Alpha objective, and learning-rate schedule differ from the combination study. Consequently, the difference between their aggregate scores is not an estimate of a history-by-future interaction. The planned matched current-pose/GT-motion and current-image/history contrasts will test whether privileged motion reduces the incremental value of visual history. Completed probe extraction alone does not answer that question.

5.5. Execution and Learning Diagnostics

Future-screen public policy calls have same-device mean times of 177.071–184.212 ms and p95 times of 178.726–185.133 ms across conditions. Peak allocated memory is 1.571–1.617 GiB. These measurements include internal perception and action computation but exclude external oracle access, I/O, rendering, loading, and priming. They come from one fixed-order profiling round; small differences do not support a stable efficiency ranking. In particular, the simulation's 50 ms action interval is not evidence of 20 Hz wall-clock deployment.

The hard-task diagnostic also identifies a supervision/execution mismatch. In the historical full-passage set, all 800 demonstrations release at the final action, with release supervision at chunk position h_7 and none at the executed h_0. This is a coverage issue under one-step deployment, rather than absence of every release label. Short diagnostic training gives inconsistent successes, and the tested first-action weighting endpoints do not establish a solution. This evidence motivates checking supervision at the executed action position before explaining difficult-task failure through future representation alone. Appendix G gives the scope and negative results.

5.6. Remaining Studies and Final Model Selection

Backbone and information flow. After completing the combination and history controls, the matched D/E study uses three fresh training seeds per backbone, followed by the seven information-flow conditions in Sec. 4. The planned paired comparisons measure auxiliary supervision, online access, and generation order; their result slots are **[D/E and R0–R6 outcomes and cost]**. A retained representation is revisited after changing information flow, and the selected flow is applied to the previously unselected backbone. Necessary multi-seed and unused-scene confirmation precedes the final design. Appendix E gives the comparisons and missing generative interface.

Robot-data pretraining and independent adaptation. The confirmed design proceeds to training-recipe, model/data-scale, and compute studies. DROID preparation must validate geometry, timing, and selected targets; it does not directly provide object pose GT. A real-robot-data base independently initializes the simulation and real-robot post-training branches. Matched downstream budgets and equally budgeted recipe searches evaluate initialization. Results are **[pending]**; Appendix F gives the protocol.

External and deployment evidence. Single-arm DOMINO, static compatibility, and real conveyor manipulation test transfer after selection, using their native scores. Measured-delay tests report contact error, safety, action age, and end-to-end cost. The final SE(3)-WAM configuration and external results remain **[pending]** (Appendix F).

6. Discussion and Conclusion

Dynamic manipulation calls for a representation of the future that supports the interaction the robot must perform. The completed study shows that adding target and scene prediction objectives does not automatically improve a motion-informed policy: sparse trajectories give modest point-estimate gains, dense 2-D displacement is adverse under the tested recipe, and difficult tasks retain shared failures. Combination and history results further show why information access, training recipe, and execution must accompany a representation comparison.

The evidence currently covers fourteen simulated tasks, one training seed for the main screens, privileged target information, and synchronous execution. The fine-target/coarse-context hypothesis, useful online foresight, multi-seed design stability, and real-robot transfer require the specified comparisons. Final selection will use task-specific gains and losses, action-level interventions, and measured deployment cost.

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